Xinyi (Rex) Le

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Education

Georgia Institute of Technology — Atlanta, GA
Doctor of Philosophy in Computer Science — *Aug 2026 – May 2031 (Expected)*

Rice University — Houston, TX
Bachelor of Science in Computer Science, Bachelor of Arts in Mathematics — *Aug 2022 – May 2026*

Relevant Coursework: Logic and CS and AI, Natural Language Processing, Deep Learning Systems, Compilers, Real/Complex Analysis, Abstract Algebra, Integration Theory.

Publications

Reviving DSP for Advanced Theorem Proving in the Era of Reasoning Models — NeurIPS 2025 (Poster)

- Built a REPL-based pipeline enabling LLM–Lean 4 interaction for proof repair and library refinement, extended to the DSP+ framework for end-to-end neuro-symbolic theorem proving.

Main Experience

Research Intern — May 2025 – Aug 2025
Harvard John A. Paulson School of Engineering and Applied Sciences — *Boston, MA*

- Code can be found here: **Library-learning link**.
- Conducted research on the Simply Typed Lambda Calculus and semantic reasoning with Language Models.
- Constructed a Python-OCaml Foreign Function Interface (FFI) with intermediate C-stubs to integrate a parser and type-checker, that further supports extraction of libraries of definitions for a library learning environment.
- Implemented and test an in-memory LRU cache for a Transformer model using Jax and Equinox that supports both retrieval and insertion/update operations.

Research Intern — May 2024 – Jan 2025
Microsoft Research Asia — *Beijing, China*

- Investigated the integration of large language models with formal proof assistants to develop neuro-symbolic systems for automated theorem proving.
- Constructed an experimental curriculum-learning pipeline, adopting a human-like chapter-by-chapter strategy of pre- and post-learning and Lean 4 library generation.
- Implemented a REPL-based pipeline enabling interaction between LLMs and Lean 4 that supports code execution, proof repair, and library refinement. It was further extended to the DSP+ framework for end-to-end neuro-symbolic reasoning.
- Designed system prompts for various functions within the framework, including OCR-based PDF recognition, auto-formalization for library generation, quiz-solution formatting, result evaluation, etc.
- This work led to a NeurIPS 2025 poster publication (see Publications section).

Research Assistant — May 2024 – Jan 2025
Department of Statistics, Rice University — *Houston, TX*

- Developed a fan-shape model to detect ventricular structures in echocardiographic images and generate corresponding segmentation masks.
- Processed and analyzed a large dataset of cardiac ultrasound videos to support model development and evaluation.

Research Intern — May 2023 – Sep 2023
Microsoft Research Asia — *Shanghai, China*

- Participated in research on in-context executing and tool-augmented reasoning of large language models.
- Designed and optimized system prompts for different LLMs, recording model outputs and labels in JSON/CSV formats; applied multi-threading to improve data labeling efficiency by over 200%.
- Implemented a logistic regression classifier on the GSM8K dataset to predict solver suitability, achieving 90% accuracy and 0.85 AUC.

Projects

InvestExpress | *Java, Spring Boot, React, Firebase Auth, Google Cloud, Bigtable* Jan 2025 - May 2025

- Implemented secure user authentication and session management using Firebase Auth and token-based verification in Spring Boot, ensuring encrypted credential handling and protected API endpoints.
- Developed backend AuthService and integrated Firebase token validation across portfolio, transaction, and stock endpoints for multi-user data isolation and privacy.
- Built the frontend login/signup flow in React, including real-time error handling, Google sign-in integration, and persistent session renewal with local storage and secure cookies.
- Designed and tested session timeout and renewal mechanisms, improving user experience and maintaining compliance with web security best practices.

ArticleHub | *React, Node.js, Express, Mongoose, Heroku, Jest, Jasmine* Aug 2024 – Dec 2024

- Code can be found here: **ArticleHub link**.
- Built a full-stack article sharing platform supporting user registration, login, Google login (Passport.js), article posting, comments, and user following/unfollowing.
- Integrated profile and avatar updates with Cloudinary.
- Deployed the backend with Express on Heroku and the frontend using React and Surge.
- Managed persistent user, article, profile data with MongoDB.

OwlDB/M3ssag1n8 | *Go, HTML/CSS, TypeScript, Cypress* Aug 2024 – Dec 2024

- Built a network accessible NoSQL database using RESTful web service in a three-member team. (OwlDB)
- Implemented concurrent skip lists with lazy synchronization and atomics for efficient upserting and querying.
- Supported real-time updates via server-sent events (SSE) for subscriptions to documents and collections.
- Built a web-based messaging application using OwlDB as the backend web service, enabling users to create, view, and manage workspaces, channels, and posts. (M3ssag1n8)

REMM Loan | *Python, Pandas, Random Forest, Scikit-learn, Cross-validation* Sep 2023

- DevPost Website: **REMM Loan link**
- Implemented a financial application to estimate a person's loan eligibility in a four-member team.
- Trained a Standard Random Forest model on 14 features from 2022 FFIEC Loan/Application Records, achieving 89% accuracy through cross-validation.

Technical Skills

Languages: Python, Java, Lean 4, Go, R, C, SQL (Postgres), HTML/CSS, JavaScript, Haskell, OCaml
Frameworks: React, Node.js, Spring Boot, JUnit, Express, Jest, Jasmine
Developer Tools: Git, VS Code, PyCharm, IntelliJ, RStudio, Heroku, Microsoft Azure, Google Cloud
Databases: MongoDB, Neo4j, Firebase
Libraries: OpenAI, Pandas, NumPy, Matplotlib, Scikit-learn, JSON/JSONLines, OpenCV, Mongoose

Teaching Experience

Teaching Assistant Aug 2025 – Dec 2025
COMP431: Web Development *Rice University*

- Collaborated with instructors and fellow TAs in weekly meetings to refine grading rubrics, ensure consistency across 7 web development assignments, and uphold best practices in full-stack design and testing.
- Held weekly office hours for a class of 60+ students, providing guidance on React component architecture, state management, and asynchronous data flow with RESTful APIs.
- Assisted students in debugging authentication workflows, database connectivity (MongoDB), and deployment issues on Heroku, reinforcing practical DevOps and security concepts.
- Provided feedback on project scalability and modularity, drawing on experience from developing ArticleHub, a full-stack article sharing platform built with React, Node.js, and MongoDB.

Teaching Assistant Jan 2026 – May 2026
COMP 440: Artificial Intelligence *Rice University*

- Lead weekly recitation sessions for a class of 80+ students, walking students through assignment solutions and core reasoning frameworks including search, adversarial reasoning, MDPs, CSPs, probabilistic models, and machine learning.
- Hold weekly office hours to answer conceptual questions and debug implementations, helping students clarify modeling assumptions and reasoning steps.
- Collaborate with instructors and fellow TAs on grading and rubric design, ensuring fair and consistent evaluation across assignments.

Honors & Awards

Successful Participant Apr 2025
The Mathematical Contest in Modeling 2025

President's Honor Roll Spring 2025, Spring 2023
Rice University

National Gold (Division 2) Mar 2021
35th Annual AAPT Physics Bowl Contest

Silver Medal Award in Energy & Environment, final round Oct 2020 - Mar 2021
2020-2021 Conrad Challenge, China FINALS

Certificate of Participation Mar 2021
AIME (American Invitational Mathematics Examination)

Gold (Global Ranking) Nov 2020
British Physics Olympiad (BPhO) Round 1, 2020–2021

Extracurricular Activities and Hobbies

Content Creator Aug 2021 – Present
Bilibili / Youtube

- Bilibili personal space: **Bilibili**. Youtube personal channel: **Youtube**.
- Analyzed Minecraft mods/modpacks, and other web-based game systems through mathematical modeling (exponential/logarithmic growth, resource optimization) and Java code inspection. Produced hundreds of commentary and strategy videos that explain in-game mechanics, derive optimal play strategies, and help players achieve efficient and enjoyable gameplay.
- Edited videos using timeline management, keyframing, beat-matching, transitions, captions, and script planning to deliver clear and engaging content.
- Established an online community through group chats, fostering player interaction, strategy discussions, and gameplay collaboration. Gained more than 1M cumulative views and 3K followers.

Family Tutor Aug 2015 – Present
Home

- Conducted weekly one-on-one teaching sessions for my middle school-aged younger sister, focusing on mathematics, foundational physics, and STEM concepts.
- Incorporated diverse activities, including Sudoku and logic puzzles, to actively stimulate critical thinking, strategic planning, and abstract problem-solving skills.
- Focused on teaching "how to think" rather than rote memorization, fostering her long-term interest in STEM fields and improving academic confidence.

Piano Aug 2011 – Present
Anywhere

- Achieved the highest amateur level (Grade 10, Shanghai Conservatory of Music).